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CBCS SCHEME

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BCS402

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Microcontroller

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain the purpose of various fields of current program status register with a neat diagram.	05	L2	CO1
	b.	Explain the ARM design philosophy.	06	L2	CO1
	c.	Explain the core extensions of ARM processor with neat block diagram.	09	L2	CO1
		OR			
Q.2	a.	Explain Embedded systems hardware with a neat block diagram.	06	L2	CO1
	b.	What is pipelines in ARM? Illustrate with an example the pipeline stage of ARM 9 and ARM 10.	09	L2	CO1
	c.	Describe the RISC design philosophy with 4 design rules.	05	L2	CO1
		Module – 2			
Q.3	a.	Explain the following with examples : (i) RSC (ii) MLA (iii) STRH (iv) SWP	10	L2	CO2
	b.	Explain the different data processing instruction in ARM.	10	L2	CO2
		OR			
Q.4	a.	Explain Barrel shifter instruction in ARM with suitable examples.	10	L2	CO2
	b.	Explain the different branch instruction of ARM processor.	05	L2	CO2
	c.	Explain co-processor instruction of ARM processor.	05	L2	CO2
		Module – 3			
Q.5	a.	Explain the different basic data types in C. Provide examples of how each data type can be used in a C program.	08	L2	CO3
	b.	Discuss the concept of register allocation in compiler optimization. Illustrate its significance with an example.	07	L2	CO3
	c.	Describe the process of a function call in C.	05	L2	CO3
		OR			
Q.6	a.	Discuss the common portability issues faced when writing C programs. How can these issues be mitigated.	07	L2	CO3
	b.	Explain the concept of pointer aliasing with example.	07	L2	CO3
	c.	How are function calls handled efficiently in calling function in C?	06	L2	CO3
		Module – 4			
Q.7	a.	What are interrupts? Discuss interrupt vector table with diagram for ARM processor.	06	L2	CO4
	b.	Describe the sequence of operations that occurs when an ARM processor handles an IRQ exceptions.	06	L2	CO4
	c.	Discuss the priority system for exception in ARM processor.	08	L2	CO4
		OR			
Q.8	a.	Explain the role of the link register in ARM exception handling.	08	L2	CO4
	b.	Explain the design and implementation of an interrupt stack in a ARM-based system. Explain the steps involved.	08	L2	CO4
	c.	What are the key differences between a boot loader and firmware?	04	L2	CO4

Module – 5					
Q.9	**a.**	Explain the basic operation of a cache controller.	**06**	**L2**	**CO5**
	b.	With a neat diagram, explain the basic architecture of a cache memory.	**10**	**L2**	**CO5**
	c.	Mention any 4 relationship between cache and main memory.	**04**	**L2**	**CO5**
OR					
Q.10	**a.**	Write a note on cache write policy both write back or write through.	**10**	**L2**	**CO5**
	b.	Describe the allocation policy on a cache miss.	**04**	**L2**	**CO5**
	c.	Write a note on following : (i) Write buffers (ii) Cache efficiency	**06**	**L2**	**CO5**

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Fourth Semester B.E. Degree Examination

Subject Title: Microcontrollers

TIME: 03 Hours **Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE.**

		Module -1	***Bloom's Taxonomy Level**	**Marks**
Q.01	a	Mention the difference between 1. Microctroller and Microprocessor 2. RISC and CISC	L2	10 Marks
	b	Explain the architecture of an ARM embedded device with the help of neat diagram.	L2	10 Marks
		OR		
Q.02	a	Explain in detail about Current Program Status Register (CPSR).	L2	10 Marks
	b	With a neat diagram, explain embedded system Hardware.	L2	10 Marks
		Module-2		
Q. 03	a	Explain different arithmetic instructions in ARM processor with an example.	L2	10 Marks
	b	Explain single register load store addressing mode syntax, table, index mode with an example.	L2	10 Marks
		OR		
Q.04	a	Explain barrel shifter instructions in ARM with suitable examples.	L2	10 Marks
	b	Explain different Logical instructions in ARM processor with an example.	L2	10 Marks
		Module-3		
Q. 05	a	Explain code optimization, profiling and cycle counting.	L3	10 Marks
	b	Write a C program that prints the square of the integers between 0 to 9 using functions and explain how to convert this C function to an assembly function with command.	L3	10 Marks
		OR		
Q. 06	a	Discuss how registers are allocated to optimize the program.	L3	10 Marks
	b	Develop an ALP to find the sum of first 10 integer numbers.	L3	10 Marks
		Module-4		
Q. 07	a	With a neat diagram explain ARM processor exceptions and modes.	L2	10 Marks
	b	Explain assigning interrupts and interrupt latency.	L2	10 Marks
		OR		
Q. 08	a	Briefly explain what happens when an IRQ and FIQ exception is raised with an ARM processor.	L2	10 Marks

	b	Explain firmware execution flow and explain Red Hat RedBoot.	L2	Marks
Module-5				
Q. 09	a	Explain the basic architecture of cache memory.	L2	10 Marks
	b	Explain how main memory maps to a cache memory.	L2	10 Marks
OR				
Q. 10	a	With a neat block diagram explain associative cache.	L2	10 Marks
	b	Briefly explain cache line replacement policies.	L2	10 Marks

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs to be attained by every bit of questions.

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Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024

Microcontrollers

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

			M	L	C
		Module – 1			
Q.1	a.	Explain the architecture of an arm embedded device with a neat diagram.	10	L2	CO1
	b.	How are monitor and control internal operations performed in ARM core? Explain in brief.	10	L2	CO1
		OR			
Q.2	a.	Explain memory management in ARM core. Compare cache and tightly coupled memory.	10	L2	CO1
	b.	Explain mechanism applied by ARM core to handle exception, interrupts using different vector table.	10	L2	CO1
		Module – 2			
Q.3	a.	Examine data processing instructions requirement in the manipulation of data register? Explain in brief data processing instructions.	10	L2	CO2
	b.	Explain with examples the following 32-bit instruction of ARN processor i) CMN ii) MLA iii) MRS iv) BIC v) LDR.	10	L2	CO2
		OR			
Q.4	a.	Explain the following with example: i) Stock operation ii) Swap instructions.	10	L2	CO2
	b.	Explain Branch instructions in ARM with suitable example. Demonstrate Branch instruct usage flow of execution with an example program.	10	L2	CO2
		Module – 3			
Q.5	a.	How registers are allocated to optimize the program? Develop an assembly level program to find the sum of first to integer numbers.	10	L2	CO3
	b.	How complier handles a "for loop" with variable number of iterations N and loop controlling with an example.	10	L2	CO3
		OR			
Q.6	a.	Explain the following terms with an appropriate example: i) Pointer Aliasing ii) Portability issues.	10	L2	CO3
	b.	How function calling is efficiently used by ARM through APCS with an example program.	10	L2	CO3
		Module – 4			
Q.7	a.	Explain ARM processors exception and modes with a neat diagram.	10	L2	CO4
	b.	Explain exception priorities and link register offset.	10	L2	CO4
		OR			
Q.8	a.	List ARM firmware suite features. Explain firmware execution flow and Red Hat Boot.	10	L2	CO4
	b.	Explain IRQ and Fir exception, also to enable and disable IRQ and FIQ interrupts.	10	L2	CO4
		Module – 5			
Q.9	a.	Explain basic architecture of cache memory.	10	L2	CO5
	b.	Explain process involved in main memory mapping to a cache memory.	10	L2	CO5
		OR			
Q.10	a.	Explain with diagram set associative cache. How are efficiency is measured?	10	L2	CO5
	b.	Briefly explain cache line replacement policies with an example.	10	L2	CO5

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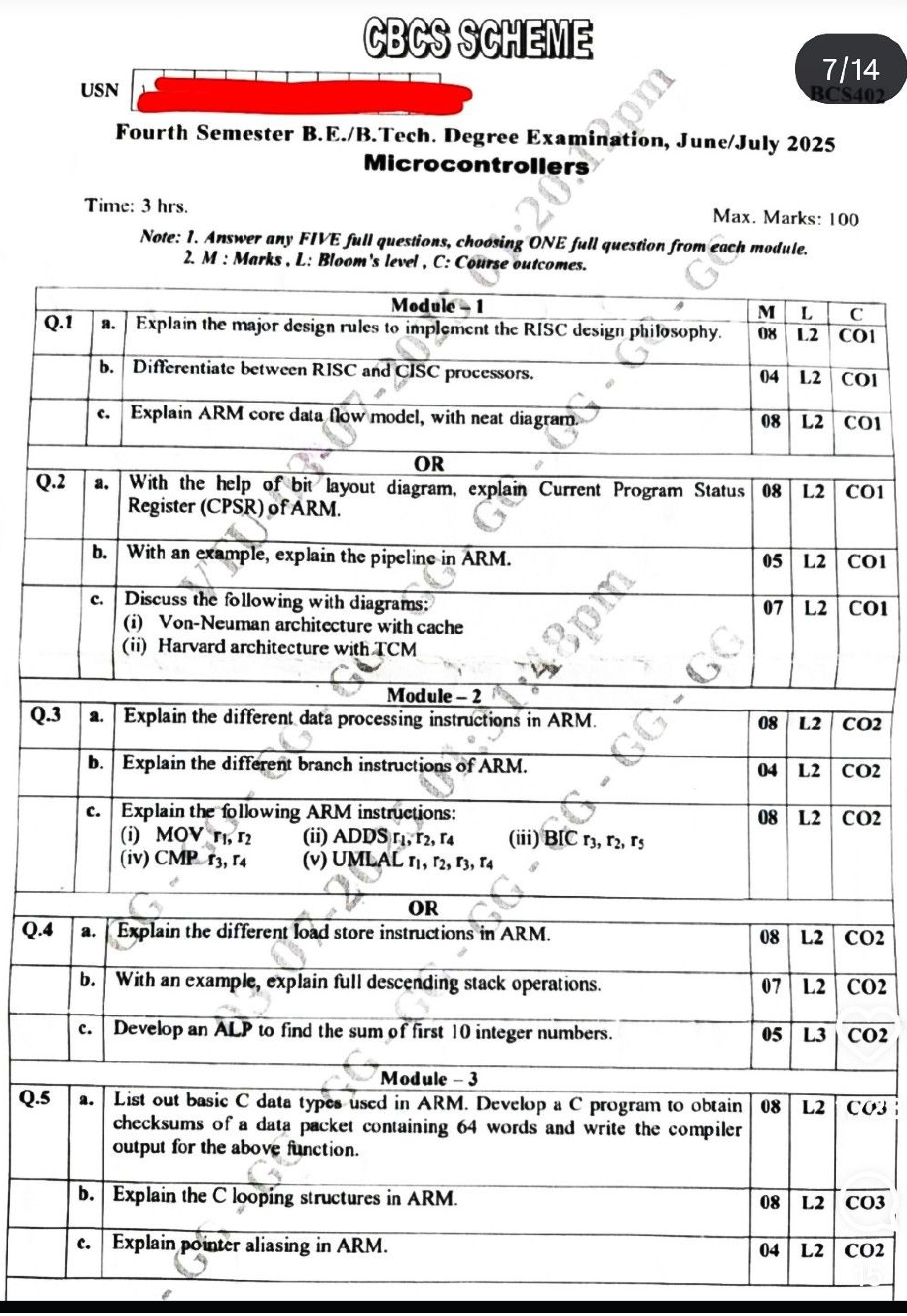

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BCS402

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2025
Microcontrollers

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

			M	L	C
		Module – 1			
Q.1	a.	Explain the major design rules to implement the RISC design philosophy.	08	L2	CO1
	b.	Differentiate between RISC and CISC processors.	04	L2	CO1
	c.	Explain ARM core data flow model, with neat diagram.	08	L2	CO1
		OR			
Q.2	a.	With the help of bit layout diagram, explain Current Program Status Register (CPSR) of ARM.	08	L2	CO1
	b.	With an example, explain the pipeline in ARM.	05	L2	CO1
	c.	Discuss the following with diagrams: (i) Von-Neuman architecture with cache (ii) Harvard architecture with TCM	07	L2	CO1
		Module – 2			
Q.3	a.	Explain the different data processing instructions in ARM.	08	L2	CO2
	b.	Explain the different branch instructions of ARM.	04	L2	CO2
	c.	Explain the following ARM instructions: (i) MOV r_1, r_2 (ii) ADDS r_1, r_2, r_4 (iii) BIC r_3, r_2, r_5 (iv) CMP r_3, r_4 (v) UMLAL r_1, r_2, r_3, r_4	08	L2	CO2
		OR			
Q.4	a.	Explain the different load store instructions in ARM.	08	L2	CO2
	b.	With an example, explain full descending stack operations.	07	L2	CO2
	c.	Develop an ALP to find the sum of first 10 integer numbers.	05	L3	CO2
		Module – 3			
Q.5	a.	List out basic C data types used in ARM. Develop a C program to obtain checksums of a data packet containing 64 words and write the compiler output for the above function.	08	L2	CO3
	b.	Explain the C looping structures in ARM.	08	L2	CO3
	c.	Explain pointer aliasing in ARM.	04	L2	CO2

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		OR			
Q.6	**a.**	With an example, explain function calls in ARM.	**08**	**L2**	**CO3**
	b.	Explain register allocation in ARM.	**07**	**L2**	**CO3**
	c.	Write a brief note on portability issues when porting C code to ARM.	**05**	**L2**	**CO3**
		Module – 4			
Q.7	**a.**	Explain the ARM processor exceptions and modes, vector table and exception priorities.	**10**	**L2**	**CO4**
	b.	Explain the interrupts in ARM.	**10**	**L2**	**CO4**
		OR			
Q.8	**a.**	Explain the ARM firmware suite and red hat redboot.	**10**	**L2**	**CO4**
	b.	Explain the sandstone directory layout and sandstone code structure.	**10**	**L2**	**CO4**
		Module – 5			
Q.9	**a.**	Explain the basic architecture of a cache memory and basic operation of a cache controller.	**10**	**L2**	**CO5**
	b.	With a neat diagram, explain a 4 KB, four way set associative cache.	**10**	**L2**	**CO5**
		OR			
Q.10	**a.**	Explain the write buffers and measuring cache efficiency.	**08**	**L2**	**CO5**
	b.	Explain the cache policy.	**12**	**L2**	**CO5**

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